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Part 2(a) Flight Delay Analysis for 2004 to 2008

This analysis analyses flight delay patterns to identify optimal travel times that minimize flight delay. Using flight data, we reveal how time of day and weekday selection significantly impact delay durations.

Methodology

Data Wrangling Process

1. Data cleaning:

We clean the data by removing entries where the departure time, departure delay and arrival delay values are missing (NA).

2. Delay adjustments:

Delays shorter than 15 minutes are treated as negligible (set to 0) and total delay is calculated as the sum of adjusted departure and arrival delays.

3. Feature adjustments:

Flights are categorized into four time periods (Early Morning, Morning, Afternoon, Night) based on departure hour and weekdays are derived from flight dates for analysis.

Analysis approach

- Group flights by time categories and weekdays
- Calculate the mean total delays for each group
- Rank periods by delay performance
- Visualize results using bar charts

Results

Optimal Flight Times

A tibble: 4 × 2

TimeCategory <chr>	AvgDelay <dbl>
Morning	10.18995
Afternoon	22.62753
Early Morning	29.30804
Night	43.87479

4 rows

Morning flights (6 - 11.59am) show the best performance with average delays of just 10.2minutes. Night flights demonstrate the worst reliability with average delays of 43.9 minutes.

Best Travel Days

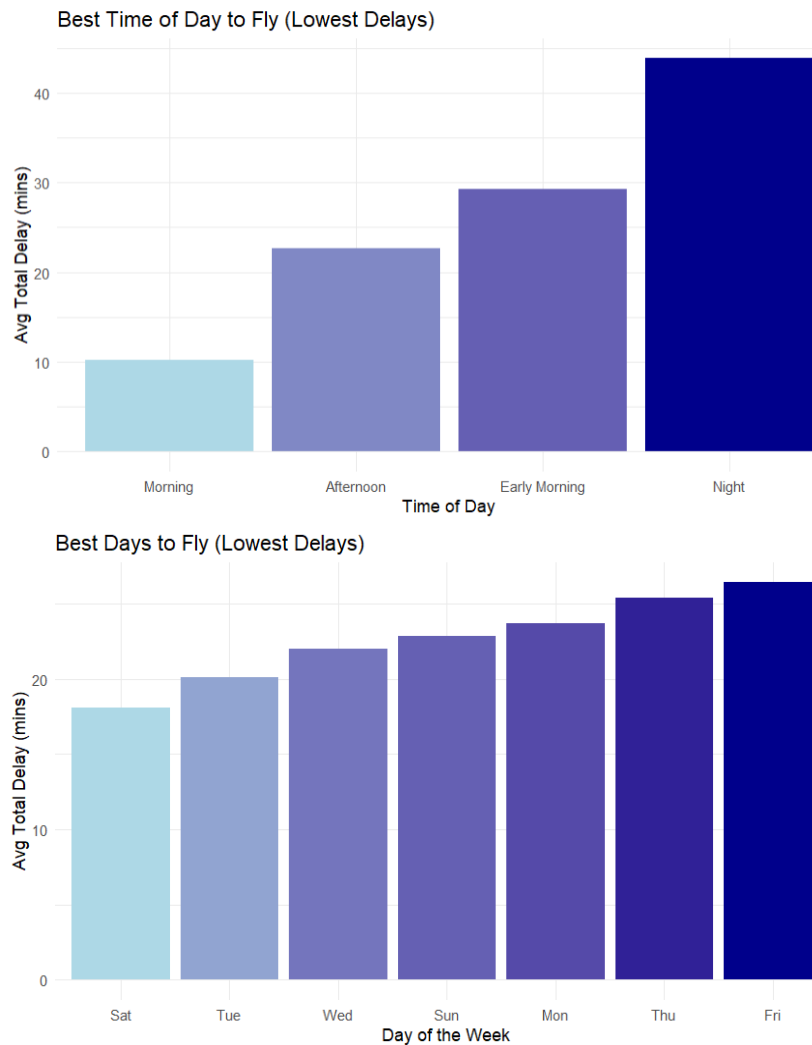
A tibble: 7 × 2

DayOfWeek <ord>	AvgDelay <dbl>
Sat	18.10468
Tue	20.07167
Wed	21.96427
Sun	22.80765
Mon	23.70672
Thu	25.36956
Fri	26.42420

7 rows

Saturday flights offer the most reliable travel with average delays of 18.1 minutes. Friday flights show the poorest performance at 26.4 minutes, likely due to increased weekend travel volume.

Visualization



Recommendations for Travelers

The optimal travel time and day is morning flights (6am-12pm) on Saturday or Tuesday. In contrast, travelers need to avoid night flights (8pm-12am) on Friday which combine the worst time and worst day factors.

Conclusion

These findings can be valuable for travelers and airlines who want to minimize delays and optimize scheduling. Travelers should consider booking flights on Saturday or Tuesday mornings for a higher likelihood of on-time departures. In addition, airlines could schedule more critical or high-priority flights during low-delay periods to improve overall performance.

Part 2(b) Aircraft Age and Flight Delays Analysis for 2004 to 2008

By analyzing five years of flight data, this analysis evaluates how aircraft age influences delays which determine whether older planes contribute to more flight delays.

Methodology

Data Wrangling Process

1. Data cleaning

The analysis begins by removing all cancelled and diverted flights from the dataset. Records with missing delay values or aircraft information are excluded to ensure data quality.

2. Key features selection

The study focuses on flight year, tail number, departure and arrival delay metrics. Aircraft manufacturing years are extracted from plane data.

3. Age calculation

Aircraft age is determined by matching flights to manufacturing years through tail numbers and then calculating the difference. Invalid ages (negative or >100 years) are excluded.

4. Delay adjustment

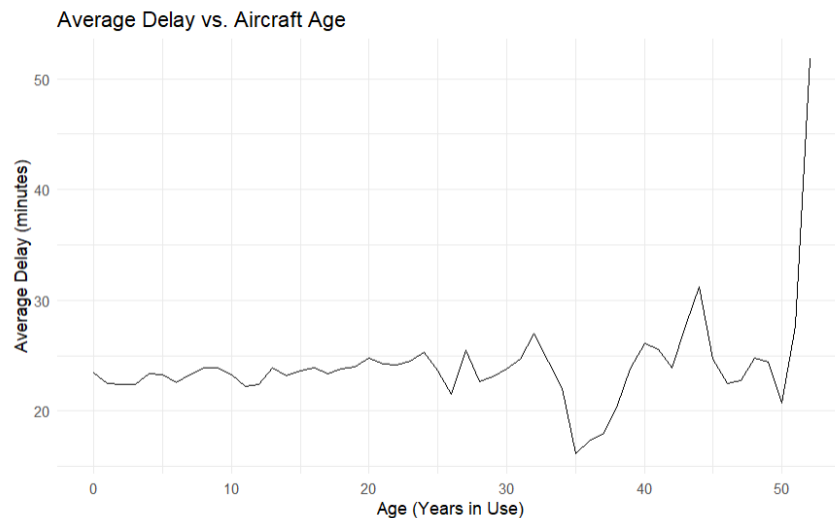
Delays under 15 minutes are ignored while longer delays are summed into total delay. Data quality is ensured by verifying distributions and filtering outliers.

Analysis Approach

- Remove cancelled flights and missing values.
- Combine departure/arrival delays, ignoring minor delays (<15 mins)
- Identify key age thresholds where delays increase.
- Plot and explain age-delay patterns for decision-making.

Result

Visualization



The graph shows the relationship between aircraft age (in years) and the average flight delay (in minutes). Key findings from the graph include:

1. Stable delay for newer aircraft: Aircraft under approximately 35 years old exhibit relatively stable average delays, mostly ranging between 20 and 30 minutes.
2. Increased variability in older aircraft: Aircraft older than 35 years start showing increased fluctuations in delay times which indicate possible reliability issues or operational inefficiencies.
3. Tremendous increase for the oldest aircraft: Aircraft over 50 years old experience a significant surge in average delay, exceeding 50 minutes. This suggests that very old aircraft may contribute disproportionately to flight delays, possibly due to maintenance concerns.

Conclusion

For airlines, retiring aircraft before 25 years and enhancing maintenance for those over 10 years can help reduce delays and operational inefficiencies. Balancing fleet age with efficiency is key to minimizing passenger disruptions.

For passengers, choosing flights on newer aircraft (under 10 years) can lower the risk of delays, while those on older planes (over 25 years) should expect potential wait times. Checking aircraft age through flight tracking services can aid in making informed travel choices.

Part 2(c) Flight Diversion Prediction Analysis for 2004 to 2008

This report investigates the factors influencing flight diversions by developing a predictive logistic regression model. Using five years of flight data, we analyze how flight times, carrier operations and flight distance affect diversion probabilities while tracking how these relationships evolve annually.

Methodology

Data Wrangling Process

- 1. Initial filtering:**
Cancelled flights are removed from the dataset, ensuring only valid operational flight data is used.
- 2. Feature selection:**
From the original 28 available features, six key features are retained based on their operational relevance and data completeness for diversion prediction modeling.
- 3. Type conversion:**
Categorical variables including DayOfWeek and UniqueCarrier are converted to factors, while Year is maintained as string for temporal analysis and the Diverted target receives binary encoding (0=normal, 1=diverted).
- 4. Time features adjustments:**
Scheduled flight times are processed into hour bins with proper time ranges (CRSDepTime: 0-2359, CRSArrTime: 0-2359).

Feature Selection through Explanatory Data Analysis (EDA)

Our exploratory analysis reveals several important patterns that guide feature selection. The visualization of coefficients over time clearly shows that flight distance consistently emerges as the strongest predictor of diversions across all years. Scheduled arrival time demonstrates increasing importance over the study period while departure time shows gradually decreasing predictive power. We exclude weather and security features despite their potential relevance because they contain excessive missing data. The day of week and carrier information are retained as they show meaningful variation in diversion rates.

Machine Learning Pipeline

The modeling approach employs a constructed logistic regression pipeline. After splitting the data while preserving temporal relationships, we implement preprocessing to handle categorical variables and normalize numerical features. The model incorporates regularization to address potential overfitting given the severe class imbalance. We train separate models each year to track how feature importance evolved over time as clearly visible in the coefficient trend plot.

Classification Error

```
classif.ce  
0.002179214
```

The model achieves a 0.22% classification error, demonstrating near-perfect accuracy in predicting flight diversions despite the rare occurrence of such events. This confirms flight distance and scheduled times are key diversion predictors, enabling airlines to proactively manage high-risk flights. While exceptionally accurate, periodic validation with new data will ensure continued reliability as operational conditions evolve.

Result

Visualization



The coefficient plot reveals three significant patterns. First, distance maintains its position as the dominant predictor throughout the entire period, though its influence slightly diminishes over time. Second, arrival time shows a steady increase in importance, likely reflecting growing airport congestion during evening hours. Third, departure time becomes progressively less significant which suggests improvements in departure management. These trends indicate that while distance remains crucial, temporal factors are becoming increasingly important in diversion prediction.

Conclusion

Distance remains the strongest diversion predictor, urging airlines to prioritize long-haul planning. The rising impact of arrival time demands better arrival management while declining departure time importance reflects operational improvements. These findings enable targeted airline strategies and smarter traveler choices.